

1. To Create Main Loop

Display Prompt

Read User Input

Handle Exit Conditions

Design Control Flow Diagram

Test Interactive Loop

2. To Capture Keyboard Input

Handle Backspace

Process Enter Key

Manage Input Buffer

Support Multi-Character Commands

Test User Interaction

Code:

```
priyanka@DESKTOP-HOISO7: ~$ nano interactive_shell.c
GNU nano 8.7.1 interactive_shell.c
/* Read one character at a time */
raw.c_cc[VMIN] = 1;
raw.c_cc[VTIME] = 0;

/* Apply new settings */
tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

/* Read keyboard input manually */
int read_input(char *buffer, int size)
{
    int position = 0;
    char ch;

    while (1)
    {
        /* Read one character */
        if (read(STDIN_FILENO, &ch, 1) != 1)
            return -1;

        /* ENTER key */
        if (ch == '\n' || ch == '\r')
        {
            buffer[position] = '\0';
            printf("\n");
            return position;
        }

        /* BACKSPACE key */
    }
}
```

```
    if (ch == 127 || ch == 8)
    {
        if (position > 0)
        {
            position--;

            /* Remove character from terminal */
            printf("\b \b");
            fflush(stdout);
        }

        continue;
    }

    /* Ctrl+D */
    if (ch == 4)
    {
        buffer[position] = '\0';
        printf("\n");
        return -2;
    }

    /* Normal printable character */
    if (ch >= 32 && ch <= 126)
    {
        if (position < size - 1)
        {
            buffer[position] = ch;
            position++;
        }
    }
}
```

```
        position++;

        /* Display typed character */
        putchar(ch);
        fflush(stdout);
    }
}

/* Process the command */
void process_command(char *command)
{
    if (strcmp(command, "") == 0)
    {
        return;
    }

    if (strcmp(command, "help") == 0)
    {
        printf("\nAvailable commands:\n");
        printf("  help  - Display available commands\n");
        printf("  hello - Display greeting\n");
        printf("  pid   - Display current process ID\n");
        printf("  exit  - Exit the program\n\n");
    }
    else if (strcmp(command, "hello") == 0)
    {
        printf("Hello! Welcome to the interactive shell.\n");
    }
}
```

```

GNU nano 8.7.1 interactive_shell.c *
}
else if (strcmp(command, "pid") == 0)
{
    printf("Process ID: %d\n", getpid());
}
else
{
    printf("Unknown command: %s\n", command);
    printf("Type 'help' for available commands.\n");
}
}

int main()
{
    char input[BUFFER_SIZE];
    int result;

    struct termios original_terminal;

    /* Enable raw keyboard input */
    enable_raw_mode(&original_terminal);

    /* Main interactive loop */
    while (1)
    {
        /* Display prompt */
        printf("myshell> ");
        fflush(stdout);

```

```

GNU nano 8.7.1 interactive_shell.c *
        /* Read user input */
        result = read_input(input, BUFFER_SIZE);

        /* Handle Ctrl+D */
        if (result == -2)
        {
            printf("Exiting...\n");
            break;
        }

        /* Handle input error */
        if (result == -1)
        {
            printf("\nInput error.\n");
            break;
        }

        /* Handle exit command */
        if (strcmp(input, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }

        /* Process entered command */
        process_command(input);
    }

    /* Restore normal terminal settings */

```

```
/* Restore normal terminal settings */
disable_raw_mode(&original_terminal);

return 0;
}
```

Output:

```
priyanka@DESKTOP-H0IS07A:~/process_lab$ gcc -Wall -Wextra interactive_shell.c -o interactive_shell
priyanka@DESKTOP-H0IS07A:~/process_lab$ ./interactive_shell
myshell> hello
Hello! Welcome to the interactive shell.
myshell> help

Available commands:
  help - Display available commands
  hello - Display greeting
  pid - Display current process ID
  exit - Exit the program

myshell> pid
Process ID: 1652
myshell> helloxxx
Unknown command: helloxxx
Type 'help' for available commands.
myshell> |
```